

Notes: Houston, Lin, and Ma (JF, 2012)

Regulatory Arbitrage and International Bank Flows

Contents

1	Big picture	2
2	Data and measurement	2
2.1	International bank flows	2
2.2	Regulation data	3
2.3	Institutional and country controls	3
2.4	Bank-level foreign affiliate data	4
3	Empirical specifications	4
3.1	Aggregate recipient-country inflow regressions	4
3.2	Aggregate source-country outflow regressions	4
3.3	Bilateral gravity model	4
3.4	Instrumental-variable approach	5
3.5	Bank-level probit model	5
4	Identification and empirical design	5
4.1	What identifies the main effects	5
4.2	Addressing omitted variables and reverse causality	6
4.3	Why institutional-quality splits matter	6
4.4	What the paper cannot fully prove	6
5	Tables and figures	7
5.1	Table I: Variable definitions and data sources	8
5.2	Table II: Summary statistics	10
5.3	Table III(A): Regulatory arbitrage and aggregate bank inflows	11
5.4	Table III(B): Regulatory arbitrage and aggregate bank outflows	12
5.5	Figures 1 and 2: Nonparametric changes in regulation and bank flows	13
5.6	Table IV: Bilateral gravity model of regulatory arbitrage	14
5.7	Table V: Instrumental-variable analyses	15
5.8	Table VI: Split-sample estimation across time and institutional quality	16
5.9	Table VII: Bank-level foreign affiliates and regulatory arbitrage	17
6	Short summary	17
7	Conclusion	18
7.1	Core conclusion	18
7.2	Economic intuition	18
7.3	Takeaway	18

1 Big picture

- **Question.** Do cross-country differences in banking regulation affect the direction and volume of international bank flows?
- **Core mechanism.** Banks headquartered in highly regulated countries may move funds, lending activity, or foreign affiliates toward countries with lighter regulation.
- **Regulatory arbitrage.** The paper interprets these flows as a form of regulatory arbitrage: banks exploit differences across national regulatory regimes.
- **Main aggregate result.** Bank inflows are lower into countries with stricter activity restrictions, capital requirements, external audit, financial statement transparency, supervisory power, and related bank regulations.
- **Main source-country result.** Bank outflows are higher from countries with stricter bank regulations.
- **Gravity result.** In bilateral source-recipient regressions, the “push” from strict source regulation and the “pull” from lax recipient regulation remain after controlling for income, population, distance, common language, institutional quality, and fixed effects.
- **Dynamic result.** Changes in regulation line up with changes in flows: recipient-country regulatory tightening is associated with lower inflow growth, while source-country regulatory tightening is associated with higher outflow growth.
- **Institutional-quality result.** Regulatory arbitrage is stronger when the recipient country is developed and has stronger property rights, creditor rights, and information sharing.
- **Bank-level result.** Large banks headquartered in highly regulated countries are more likely to establish foreign subsidiaries or branches in less regulated countries.
- **Policy implication.** Strict domestic regulation can be undermined if banks can shift activity across borders. This raises the value of international regulatory coordination.
- **Important qualification.** Lax regulation alone does not guarantee inflows. Strong legal and institutional environments are still important for attracting bank capital.

2 Data and measurement

2.1 International bank flows

- The main flow data come from BIS consolidated banking statistics.
- The sample follows bank claims from **26 source countries** to **120 recipient countries**.
- The main panel covers **1996–2007**.
- Foreign claims include financial assets such as loans, debt securities, properties, equities, and equity stakes in subsidiaries.
- Since BIS data report claims rather than direct flow measures, the paper constructs bank flows from changes in log foreign claims.

The bilateral bank-flow variable is

$$\text{BankFlow}_{s,r,t} = 100 \times \Delta \ln(FC_{s,r,t}),$$

where $FC_{s,r,t}$ is the foreign claims of banks headquartered in source country s on residents of recipient country r .

Aggregate recipient-country inflows are based on

$$100 \times \Delta \ln \left(\sum_s FC_{s,r,t} \right),$$

and aggregate source-country outflows are based on

$$100 \times \Delta \ln \left(\sum_r FC_{s,r,t} \right).$$

Interpretation: The key dependent variable is a growth rate of cross-border bank claims. The paper studies whether this growth is systematically related to differences in regulation.

2.2 Regulation data

- Regulation measures come from the Barth, Caprio, and Levine bank regulation surveys.
- The paper uses several survey waves to construct time-varying regulatory measures.
- For 1996–1999, regulation is based on the first survey.
- For 2000–2003, regulation is based on the second survey.
- For 2004–2007, regulation is based on the third survey.

Main regulation variables include:

- overall activity restrictions;
- restrictions on banks owning nonfinancial firms;
- capital regulatory index;
- strength of external audit;
- financial statement transparency;
- independence of supervisory authority;
- official supervisory power;
- loan classification leniency.

Interpretation: The variables capture different channels through which bank regulation can restrict or discipline bank activity.

2.3 Institutional and country controls

The empirical specifications control for:

- creditor rights;
- depth of credit information;
- property rights;
- bank concentration;
- government bank ownership;
- income and population;
- financial liberalization;

- growth opportunities;
- distance and common language in bilateral models.

Interpretation: These controls are important because regulatory gaps may be correlated with economic development, legal quality, and international integration.

2.4 Bank-level foreign affiliate data

- The paper constructs a bank-level database using BankScope.
- The bank-level sample covers **301 large banks** with headquarters in the 26 source countries.
- The foreign-presence variables record whether each bank has a subsidiary or branch in each of 120 recipient countries.
- The foreign-affiliate data are centered on 2007–2008.

Interpretation: Aggregate flows show where bank capital moves. Bank-level affiliate data show whether banks physically expand into less regulated host countries.

3 Empirical specifications

3.1 Aggregate recipient-country inflow regressions

A simplified recipient-country specification is:

$$\text{BankFlow}_{r,t} = \alpha + \beta \text{Regulation}_{r,t} + \gamma' X_{r,t} + \eta_r + \mu_t + \varepsilon_{r,t}.$$

- $\text{BankFlow}_{r,t}$ is aggregate bank inflow growth into recipient country r .
- $\text{Regulation}_{r,t}$ is a recipient-country regulatory measure.
- $X_{r,t}$ includes country-level controls.
- η_r are recipient-country fixed effects.
- μ_t are year fixed effects.

Interpretation: A negative β means that stricter regulation in the recipient country discourages bank inflows.

3.2 Aggregate source-country outflow regressions

A simplified source-country specification is:

$$\text{BankFlow}_{s,t} = \alpha + \beta \text{Regulation}_{s,t} + \gamma' X_{s,t} + \eta_s + \mu_t + \varepsilon_{s,t}.$$

Interpretation: A positive β means that stricter regulation at home pushes banks to send more capital abroad.

3.3 Bilateral gravity model

The paper combines push and pull effects in a bilateral model:

$$\begin{aligned} \text{BankFlow}_{s,r,t} = & \alpha_0 + \alpha_1 \text{Regulation}_{s,t} + \alpha_2 \text{Regulation}_{r,t} + \beta_1 \text{InstitutionalQuality}_{s,t} + \beta_2 \text{InstitutionalQuality}_{r,t} \\ & + \gamma_1 X_{s,t} + \gamma_2 X_{r,t} + \theta_1 \ln(\text{Distance}_{s,r}) + \theta_2 \text{CommonLanguage}_{s,r} \\ & + \eta_s + \varphi_r + \mu_t + \varepsilon_{s,r,t}. \end{aligned}$$

Interpretation: The key prediction is $\alpha_1 > 0$ and $\alpha_2 < 0$: capital flows from more regulated source countries toward less regulated recipient countries.

3.4 Instrumental-variable approach

The paper also estimates instrumental-variable specifications. Instruments include:

- experience of banking supervisors;
- central bank as bank regulator;
- Gini coefficients;
- years since independence;
- ethnic fractionalization;
- latitude;
- the average regulatory level of other countries in the sample.

Interpretation: IV analysis is designed to address reverse causality: regulatory choices might respond to flows rather than simply cause flows.

3.5 Bank-level probit model

For bank i from source country s and recipient country r , the bank-level model is:

$$Pr(Y_{i,s,r} = 1) = \Phi\left(\text{Regulation}_s, \text{Regulation}_r, \text{InstitutionalQuality}_s, \text{InstitutionalQuality}_r, \text{BankCharacteristics}_i, \text{CountryCharacteristics}_r\right)$$

- $Y_{i,s,r} = 1$ if bank i has a foreign subsidiary or branch in country r .
- In some specifications, $Y_{i,s,r} = 1$ only for foreign subsidiaries.
- $\Phi(\cdot)$ is the standard normal cdf.

Interpretation: The probit tests whether regulatory arbitrage affects the location of bank affiliates, not just the volume of cross-border claims.

4 Identification and empirical design

4.1 What identifies the main effects

- Cross-country variation in regulation creates differences in regulatory burden.
- Time variation from survey waves allows the paper to study changes in regulation.
- Source and recipient fixed effects absorb time-invariant country heterogeneity.
- Year fixed effects absorb global shocks to banking flows.
- Bilateral controls absorb gravity-model determinants of cross-border lending.

Interpretation: The main identifying idea is that after controlling for economic, legal, institutional, geographic, and fixed-effect differences, regulatory gaps should still predict where banks send capital if regulatory arbitrage is active.

4.2 Addressing omitted variables and reverse causality

The paper uses several strategies:

- lagged regulatory survey measures;
- country fixed effects;
- time fixed effects;
- controls for financial liberalization and growth opportunities;
- change-in-regulation analyses;
- instrumental-variable specifications;
- bank-level foreign-affiliate tests.

Interpretation: No single design fully eliminates endogeneity. The paper's credibility comes from the consistency of evidence across different specifications and levels of analysis.

4.3 Why institutional-quality splits matter

- A country with lax regulation may still fail to attract bank capital if property rights are weak.
- Strong creditor rights and information sharing reduce enforcement and screening frictions.
- Developed economies may be more attractive because they provide scale, legal certainty, and institutional stability.
- The paper therefore splits samples by development, creditor rights, information sharing, property rights, and overall institutional quality.

Interpretation: Regulatory arbitrage does not mean banks blindly chase the weakest regulator. They seek lightly regulated destinations where the legal environment is still credible.

4.4 What the paper cannot fully prove

- Regulation is not randomly assigned.
- Regulations may be changed in response to financial development or capital flows.
- BIS foreign-claims data do not perfectly separate all channels of bank activity.
- The regulatory survey variables are measured only at discrete survey dates.
- The empirical results identify strong associations, not a randomized causal effect.

Interpretation: The paper is best read as broad international evidence that regulatory gaps are economically meaningful, rather than as a laboratory experiment.

Table I
Variable Definitions and Data Sources

Variable	Definition	Original Sources
Bank flows	This variable captures the bank capital inflows from banks located in source country s to all sectors of the economy in recipient country r from year $t - 1$ to year t . It is calculated as 100 times the log difference (i.e., difference in log from $t - 1$ to t) of total foreign claims ($FCsr$) from source country s to recipient country r , that is, $100 \times \Delta \ln(FCsr)$. Foreign claims are determined by the nationality of the headquarters of the reporting bank published in the consolidated banking statistics (Table 9B) in each issue of the BIS Quarterly Review.	BIS (2008)
Creditor rights	An index aggregating creditor rights. Specifically, it indicates (1) whether restrictions, such as creditors' consent or minimum dividend, are imposed when a creditor files for reorganization (Restrictions on Reorganization); (2) whether there is no automatic stay or asset freeze imposed by the court to limit a creditor's ability to seize collateral (No Automatic Stay); (3) whether secured creditors are ranked number one in the distribution of proceeds in the liquidation of a bankrupt firm (Secured Creditor Paid First); and (4) whether existing management does not remain in control of the firm during the reorganization (No Management Stay).	La Porta et al. (1998), Djankov, McLeish, and Shleifer (2007)
Depth of credit information	An index that captures the information content of the credit information. A value of one is added to the index when a country's information sharing database contains each of the following characteristics: (1) both positive and negative credit information are available; (2) data on both firms and individual borrowers are reported; (3) data from retailers, trade creditors, utility companies, and banks are included; (4) more than two years of historical data are available; (5) data are compiled on all loans of value above 1% of income per capita; and (6) laws provide for borrowers' right to check their own data. The index ranges from 0 to 6, with higher values indicating the availability of more credit information, from either a public registry or a private bureau. If a country has no public registry or private bureau in a specific year, the index takes on the value 0. If an information agency exists in a country, the index takes on the value in that year or the value in the earliest available year.	Djankov, McLeish, and Shleifer (2007), World Bank "Doing Business" database
Top five bank concentration (all banks)	The fraction of total assets held by the five largest banks in the country.	BankScope

(continued)

5 Tables and figures

The visuals below are directly extracted from the paper and arranged compactly so adjacent figures and tables can be compared quickly.

5.1 Table I: Variable definitions and data sources

Table I—Continued		
Variable	Definition	Original Sources
Government bank ownership	The fraction of the banking system's assets in banks that are 50% or more owned by government.	BCL (2006, 2008)
Overall activity restrictions	The extent to which banks may engage in (1) underwriting, brokering, and dealing in securities, and all aspects of the mutual fund industry; (2) insurance underwriting and selling; and (3) real estate investment and development. Unrestricted = 1: a bank can conduct the full range of activities directly; Permitted = 2: full range of activities can be conducted, but some or all must be conducted in bank subsidiaries; Restricted = 3: part of the activities can be conducted; and Prohibited = 4: neither the bank nor subsidiaries can conduct the activities. Higher values indicate greater restrictiveness.	BCL (2006, 2008)
Restriction on banks owning nonfinancial firms	The extent to which banks may own and control nonfinancial firms (a higher value means more restrictive). Unrestricted = 1: a bank may own 100% shares of any nonfinancial firm; Permitted = 2: a bank may own 100% shares of a nonfinancial firm, depending on its equity capital; Restricted = 3: a bank can only own less than 100% shares in a nonfinancial firm; and Prohibited = 4: a bank cannot make any equity investment in a nonfinancial firm.	BCL (2006, 2008)
Independence of supervisory authority: overall	The degree to which the supervisory authority is independent of the government and legally protected from the banking industry. The indicator is constructed based on the following three questions. (1) Are the supervisory bodies responsible to (a) the Prime Minister, (b) the Finance Minister or other senior government officials, or (c) a legislative body (yes = 1)? (2) Whether the supervisors can be sued if they take actions against a bank (No = 1)? (3) Does the chair of the supervisory agency have a fixed term contract and how long? (= 1 if term ≥ 4). A higher value means a more independent supervisory agency.	BCL (2006, 2008)
Official supervisory power	An index aggregating supervisory power. Specifically, it indicates whether the supervisory agency has the legal right to meet directly with external auditors to discuss their report without getting approval from the bank; receive direct report from the external auditor on any presumed involvement of bank management in various types of misconducts; take actions against	BCL (2006, 2008)
(continued)		

Table I, page 2

Table I—Continued		
Variable	Definition	Original Sources
	external auditors for negligence; change a bank's internal organizational structure; get access to the information on off-balance-sheet items; require the bank management to constitute provisions to cover actual or potential losses; suspend the board decision to distribute dividends, bonuses, and management fees; declare a bank's insolvency; intervene the ownership rights in a problem bank; supersede shareholder rights; replace management and directors.	
Loan classification leniency	The index is constructed based on the sum of the days a loan in arrears must be classified as substandard, doubtful, and loss. A higher value indicates less stringency.	BCL (2006, 2008)
Capital regulatory index (total)	The sum of overall capital regulatory stringency and initial capital stringency, which measures whether certain funds may be used to initially capitalize a bank and whether they are officially verified. A higher value indicates greater stringency.	BCL (2006, 2008)
Strength of external audit	The effectiveness of external audits of banks. The indicator is constructed based on whether an external audit is required for banks, whether there are specific guidelines for the extent or nature of the audit, whether the auditors are licensed or certified, whether the supervisors have the rights to get access and discuss the auditor's report without the approval of the bank, and whether the auditors are required to report directly to the supervisory agency on any presumed misconducts of bank management and are legally liable for negligence. A higher value indicates better strength of external audit.	BCL (2006, 2008)
Financial statement transparency	An index based on whether accrued interest and principal enter the income statement, whether financial institutions are required to produce consolidated financial statements, whether off-balance-sheet items are publicly disclosed, whether banks are required to disclose their risk management practices to the public, and whether bank directors are legally liable for financial misreporting. The index ranges from 0 to 6, with a higher value indicating better financial statement transparency.	BCL (2006, 2008)
Number of days to enforce contracts	Number of days counted from the lawsuit filing in court until payment (including both the days when actions take place and waiting periods in between).	Djankov et al. (2003), World Bank "Doing Business" database
(continued)		

Table I, page 3

Table I—Continued

Variable	Definition	Original Sources
Property rights	Countries with more secure property rights and legal institutions that are more supportive of the rule of law receive higher ratings.	Fraser Institute Website (2008)
Log income	Log real GDP per capita, in U.S. dollars.	World Development Indicators
Log population	Log population (millions).	World Development Indicators
Common language	Dummy variable that equals one if the two countries share a common language.	Rose (2004)
Log distance	Log geographic distance.	World Development Indicators
Financial liberalization index	An index of financial liberalization over the 1973 to 2005 period for 91 economies. The index contains various dimensions including credit controls and reserve requirements, interest rate liberalization, entry barriers, capital account restrictions, privatization, and securities market policies and supervision. Higher value indicates a higher degree of financial liberalization.	Abiad, Detragiache, and Tresselt (2010)
Growth opportunities	Following Bekaert et al., (2007), annual measure is constructed as the three-digit SIC industry composition for each country by output share according to UNIDO Industrial Statistics Database. For each SIC code, the price-earnings (PE) ratio for that industry at the global level is used to construct an implied measure of growth opportunities for each country by weighting each global industry PE ratio by its relative share for that country. This measure then is subtracted from the overall world market PE ratio to remove world discount rate effects (and is also subtracted from a five-year moving average). The difference is "growth opportunities" (LGO MA), that is, $LGO MA_{i,t} = LGO_{i,t} - \sum_{s=105} LGO_{i,t-s}$, where $LGO_{i,t} = \ln[(IPE_i W_{i,t})/(IPE_w W_t)]$, IPE_i is a vector of global industry PE ratios, $W_{i,t}$ is a vector of country-specific industry weights, and W_t is a vector of world industry weights.	Bekaert et al. (2007), Datastream, and UNIDO Industrial Statistics Database
Current account /GNP (%)	The current account divided by GNP (%).	World Development Indicators
Saving /GNP (%)	The gross savings of both the public and the private sectors divided by GNP (%).	World Development Indicators

(continued)

Table I, page 4

Read:

- The abstract states the main finding: banks transfer funds to markets with fewer regulations.
- Table I defines bank flows, institutional variables, country controls, and regulatory variables.
- The dependent variable is based on the log change in BIS foreign claims.
- The table also documents the paper's original data sources, including BIS, Barth-Caprio-Levine surveys, BankScope, World Bank, La Porta et al., and other institutional datasets.

Economic message: Table I is the measurement backbone of the paper. It clarifies how regulation, institutional quality, and bank flows are operationalized.

Table I—Continued

Variable	Definition	Original Sources
Foreign subsidiary dummy	A dummy that takes the value of one if the bank with headquarters in country s has a foreign subsidiary in country r , and zero otherwise, in 2008.	BankScope
Foreign branch/subsidiary dummy	A dummy that takes the value of one if the bank with headquarters in country s has either a foreign subsidiary or a foreign branch in country r , and zero otherwise, in 2008.	BankScope
Bank size	Log of bank total assets (thousands of USD), three-year average over 2005 to 2007.	BankScope
Bank net income	Bank's net income divided by total assets (%), three-year average over 2005 to 2007.	BankScope

B. International Bank Flows

The international bank flow measure is the main dependent variable in our analysis. This variable captures bank capital inflows from banks located in source country s to all sectors of the economy in recipient country r from year $t - 1$ to t . As there is no measure of bank flows in the BIS data, we construct such a measure by calculating the annual percentage change in total foreign claims for each source-recipient combination. More specifically, our main dependent variable is defined as 100 times the log-difference of total foreign claims ($FCsr$) from source country s to recipient country r , that is, $100 \times \Delta \ln(FCsr)$. After merging different databases and deleting missing observations, our sample covers international bank flows from 26 source countries¹¹ to 120 recipient countries from 1996 to 2007. We construct the annual bank flow variable by using the stock data on December of each year in the sample period to match the annual frequency of the other explanatory variables.¹² As can be seen from Table II, the sample mean of bank flow is 2.89, which suggests that the average bank flow from a source country to a recipient country in our sample increases by 2.9% a year over the sample period. The standard deviation of bank flows is 11.32, suggesting great variation in international bank flows over time and across countries.

C. Bank Regulation and Supervision

We use a set of variables from the three worldwide surveys conducted by Barth, Caprio, and Levine over the past decade (BCL (2006, 2008)) to measure

¹¹The 26 source countries/regions with available BIS bank flow data are: Australia, Austria, Belgium, Brazil, Canada, Chile, Denmark, Finland, France, Germany, Greece, Ireland, Italy, Japan, Mexico, the Netherlands, Norway, Panama, Portugal, Spain, Sweden, Switzerland, Taiwan, Turkey, the United Kingdom, and the United States.

¹²We reduce the impact of outliers by trimming the bank flows outside the range between -100% and +100%. This filter reduces our sample size by about 6%.

Table I, page 5

5.2 Table II: Summary statistics

Table II
Summary Statistics

This table shows the summary statistics of the key variables. For Panels A, B, and C, the sample period is 1996 to 2007. For Panel D, the sample period is 2007 to 2008.

Panel A: Bilateral Relation Variables							
Variable	Mean	Median	Std. Dev.	Min	Max	Obs.	Number of countries
Bank flows	2.89	2.03	11.32	-99.97	99.98	24,233	120
Common language	0.14	0	0.35	0	1	2,065	120
Log distance	8.03	8.33	0.92	4.80	9.90	2,065	120
Panel B: Country Level Variables							
Creditor rights	1.81	2	1.13	0	4	1,368	120
Depth of credit information	3.06	3	2.21	0	6	1,368	120
Top five bank concentration	0.80	0.83	0.18	0.19	1	1,368	120
Government bank ownership	0.20	0.12	0.24	0	0.94	1,272	113
Number of days to enforce contracts (log)	6.39	6.24	5.65	4.79	7.29	1,368	120
Property rights	5.37	5.30	1.77	1.43	9.62	1,368	120
Log income	7.50	7.43	1.59	4.41	10.63	1,368	120
Log population	2.66	2.39	1.35	0.47	7.19	1,368	120
Financial liberalization index	15.31	15.25	3.73	3.75	21.00	792	86
Growth opportunity	0.01	0.01	0.09	-0.33	0.33	1,059	92
Current account /GNP (%)	-2.28	-2.65	8.20	-44.73	55.53	1,202	104
Saving /GNP (%)	19.72	18.83	9.07	0.41	58.36	1,202	104
Panel C: Regulatory Variables							
Overall activities restrictions	7.64	8	2.03	3	12	1,272	113
Restriction on banks owning nonfinancial firms	2.57	3	0.79	1	4	1,272	113
Capital regulatory index (total)	6.00	6	1.78	1	10	1,260	111
Independence of supervisory authority: overall	1.70	2	0.89	0	3	1,236	109
Official supervisory power	11.16	11	2.44	4	16	1,296	114
Loan classification leniency (log)	7.18	6.70	1.29	3.40	9.30	1,296	114

(continued)

Table II, page 1: bilateral, country, and regulatory variables

Read:

- Mean bank-flow growth is 2.89, with standard deviation 11.32.
- The sample contains 24,233 bilateral bank-flow observations.
- Regulatory variables vary meaningfully across countries: overall activity restrictions average 7.64, capital regulation averages 6.00, and official supervisory power averages 11.16.
- The bank-level panel contains more than 35,000 bank-recipient observations.
- Foreign subsidiary is rare on average, around 6%, while broader foreign presence averages about 8%.

Economic message: The summary statistics show enough cross-country and bank-level variation for the regulatory-arbitrage tests to be meaningful.

Table II—Continued

Panel C: Regulatory Variables							
Variable	Mean	Median	Std. Dev.	Min	Max	Obs.	Number of countries
Strength of external audit	5.19	6	1.19	2	7	1,272	113
Financial statement transparency	4.01	5	0.97	1	6	1,224	108
Panel D: Cross-Section Data for Banks' Foreign Affiliates							
No. of banks							
Bank level variables							
Foreign subsidiary dummy	0.06	0	0.22	0	1	35,819	301
Foreign branch/subsidiary dummy	0.07	0	0.23	0	1	35,819	301
Bank size	18.67	18.44	1.27	17.04	21.62	35,819	301
Bank net income	0.91	0.76	0.93	-0.487	8.90	35,819	301

various aspects of bank regulation and supervision across countries. These variables include two measures of restrictions on activities (*Activities Restrictiveness* and *Bank Own Non-financial Firms*) and a measure related to capital Table II, page 2: cross-section data for bank foreign affiliates

5.3 Table III(A): Regulatory arbitrage and aggregate bank inflows

Table III (A)
Regulatory Arbitrage and Aggregate Bank Inflows

The dependent variable is aggregate bank inflows to 120 recipient countries, which is defined as 100 times the log-difference of total foreign claims (FCr) of 26 source countries to recipient country r , that is, $100 \times \Delta \ln(\Sigma \text{FCr}_r)$. For columns (1) to (7) the estimation is based on fixed effect OLS regressions. For column (8), it is based on GDP (in USD)-weighted OLS estimation. The country-level banking regulatory variables are time varying and are based on three major surveys spanning almost a decade by the World Bank (Barth, Caprio, and Levine (2008)). The values of the regulatory variables for the period 1996 to 1999 are taken from the first survey recorded in 1998/1999, for the period 2000 to 2003 are taken from the second survey that assesses the state of regulation as of the end of 2002, and for the period 2004 to 2007 are taken from the third survey that characterizes the environment as of the end of 2005. Detailed variable definitions can be found in Table 1. Time-fixed effects and recipient country-specific effects are included in the regressions but not reported. p -values are computed using heteroskedasticity-robust standard errors clustered for recipient countries and are presented in brackets. *, **, and *** represent statistical significance at the 10%, 5%, and 1% level, respectively.

	1	2	3	4	5	6	7	8
Overall activity restrictions (recipient)	-0.29 [0.015]**				-0.39 [0.035]**	-0.55 [0.021]**	-0.71 [0.014]**	
Restriction on banks owning nonfinancial firms (recipient)	-0.86 [0.029]**				-0.88 [0.171]	-1.26 [0.281]	-1.70 [0.216]	
Capital regulatory index (recipient)		-0.39 [0.086]*			-0.27 [0.020]**	-0.31 [0.073]*	-0.38 [0.058]*	
Strength of external audit (recipient)			-0.83 [0.054]*		-1.48 [0.033]**	-1.81 [0.014]**	-2.32 [0.009]**	
Fin statement transparency (recipient)					-1.27 [0.025]**	-0.95 [0.073]*	-1.98 [0.057]*	-1.98 [0.045]**

(continued)

Table III(A), page 1

Table III (A)—Continued

	1	2	3	4	5	6	7	8
Growth opportunities (recipient)							1.76 [0.017]**	2.06 [0.016]**
Sample period			1996 to 2007				1996 to 2005	
Recipient country fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Time fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	1,372	1,264	1,264	1,228	1,240	1,168	642	642
Adj. R ²	0.15	0.16	0.16	0.16	0.17	0.18	0.38	0.43
Number of recipient countries	120	111	111	108	109	103	71	71

Table III(A), page 3

Table III (A)—Continued

	1	2	3	4	5	6	7	8
Independence of supervisory authority (recipient)					-1.25 [0.635]**	-1.33 [0.632]**	-1.10 [0.629]**	-0.85 [0.632]**
Official supervisory power (recipient)					-0.24 [0.184]	-0.23 [0.074]*	-0.36 [0.025]**	-0.47 [0.020]**
Loan classification leniency (recipient)						0.69 [0.022]**	0.59 [0.016]**	0.51 [0.012]**
Creditor rights (recipient)	5.83 [0.063]*	6.11 [0.033]**	6.22 [0.032]**	5.96 [0.073]*	5.98 [0.034]**	6.93 [0.027]**	7.03 [0.032]**	7.89 [0.025]**
Info sharing (recipient)	2.42 [0.028]**	2.30 [0.081]*	2.45 [0.026]**	2.19 [0.091]*	2.20 [0.032]**	2.16 [0.085]*	2.72 [0.067]*	1.58 [0.088]*
Number of days to enforce contracts (recipient)						-0.11 [0.026]**	-0.12 [0.038]**	-0.10 [0.044]**
Property rights (recipient)	3.54 [0.014]**	3.66 [0.027]**	3.76 [0.022]**	3.31 [0.052]*	3.39 [0.064]*	3.66 [0.033]**	2.34 [0.028]**	2.81 [0.021]**
Log income (recipient)	2.92 [0.082]*	3.10 [0.018]**	2.77 [0.063]**	3.15 [0.012]**	3.10 [0.024]**	2.35 [0.088]**	3.90 [0.075]*	4.77 [0.056]**
Log population (recipient)	2.23 [0.303]	3.01 [0.256]	2.35 [0.239]	2.38 [0.277]	3.13 [0.239]	3.40 [0.043]**	3.66 [0.047]**	4.52 [0.036]**
Fin liberalization (recipient)							0.91 [0.039]**	0.97 [0.034]**

(continued)

Table III(A), page 2

Table III (B)
Regulatory Arbitrage and Aggregate Bank Outflows

The dependent variable is aggregate bank outflows from 26 source countries, which is defined as 100 times the log-difference of total foreign claims (FCs) of source country s to 120 recipient countries, that is, $100 \times \Delta \ln(\Sigma \text{FCs}_s)$. For columns (1) to (7) the estimation is based on fixed effect OLS regressions. For column (8), it is based on GDP (in USD)-weighted OLS estimation. The country-level banking regulatory variables are time varying and are based on three major surveys spanning almost a decade by the World Bank (Barth, Caprio, and Levine (2008)). The values of the regulatory variables for the period 1996 to 1999 are taken from the first survey recorded in 1998/1999, for the period 2000 to 2003 are taken from the second survey that assesses the state of regulation as of the end of 2002, and for the period 2004 to 2007 are taken from the third survey that characterizes the environment as of the end of 2005. Detailed variable definitions can be found in Table 1. Time fixed effects and source country-specific effects are included in the regressions but not reported. p -values are computed using heteroskedasticity-robust standard errors clustered for source countries and are presented in brackets. *, **, and *** represent statistical significance at the 10%, 5%, and 1% level, respectively.

	1	2	3	4	5	6	7	8
Overall activity restrictions (source)		0.53 [0.025]**				0.92 [0.024]**	1.06 [0.017]**	1.16 [0.012]**
Restriction on banks owning nonfinancial firms (source)		2.33 [0.087]*				2.28 [0.017]**	2.19 [0.014]**	1.50 [0.017]**
Capital regulatory index (source)			0.28 [0.038]**			0.41 [0.097]*	0.65 [0.076]**	0.78 [0.057]**
Strength of external audit (source)				0.85 [0.032]**		0.74 [0.218]	1.27 [0.139]	1.72 [0.113]
Fin statement transparency (source)				2.62 [0.026]**		2.74 [0.030]**	2.45 [0.026]**	2.91 [0.021]**
Independence of supervisory authority (source)					1.05 [0.083]*	1.41 [0.542]	1.89 [0.389]	1.26 [0.446]
Official supervisory power (source)						1.83 [0.131]	1.76 [0.041]**	1.28 [0.059]*

(continued)

Table III(B), page 1

Read:

- Recipient-country activity restrictions are negatively related to aggregate bank inflows.
- Restrictions on banks owning nonfinancial firms, capital regulatory stringency, external audit, financial statement transparency, supervisory independence, and supervisory power generally reduce inflows.
- Results remain when controls, fixed effects, and weighted OLS specifications are added.

Economic message: Stricter regulation in the recipient country discourages foreign bank capital. This is the “pull” side of regulatory arbitrage.

5.4 Table III(B): Regulatory arbitrage and aggregate bank outflows

Table III (B)—Continued								
	1	2	3	4	5	6	7	8
Loan classification leniency (source)						−0.44 [0.013]**	−0.32 [0.018]**	−0.25 [0.025]**
Creditor rights (source)	−3.24 [0.016]**	−3.86 [0.022]**	−3.31 [0.052]*	−2.95 [0.141]	−2.90 [0.113]	−2.89 [0.024]**	−2.77 [0.034]**	−3.09 [0.026]**
Info sharing (source)	−1.55 [0.028]**	−1.18 [0.030]**	−1.56 [0.028]**	−1.08 [0.030]**	−0.84 [0.173]	−0.76 [0.003]**	−0.97 [0.015]**	−0.70 [0.0061]**
Number of days to enforce contracts (source)						0.21 [0.014]**	0.19 [0.017]**	0.22 [0.016]**
Property rights (source)	−4.61 [0.035]**	−4.26 [0.063]*	−4.62 [0.058]*	−4.36 [0.062]*	−5.70 [0.037]**	−4.79 [0.035]**	−5.17 [0.024]**	−6.51 [0.018]**
Log income (source)	−1.02 [0.333]	−1.70 [0.032]**	−1.56 [0.330]	−1.65 [0.281]	−1.74 [0.036]**	−1.31 [0.277]	−1.82 [0.145]	−2.09 [0.108]
Log population (source)	−2.01 [0.122]	−2.68 [0.031]**	−2.13 [0.120]	−1.85 [0.124]	−2.53 [0.081]*	−2.73 [0.146]	−2.98 [0.123]	−1.71 [0.149]
Financial liberalization (source)							0.42 [0.176]	0.34 [0.228]
Growth opportunities (source)							−1.49 [0.025]**	−1.26 [0.030]**
Sample period	1996 to 2007						1996 to 2005	
Source country-fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Time-fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	238	238	238	238	238	238	181	181
Adj. R^2	0.30	0.31	0.32	0.30	0.32	0.36	0.37	0.41
Number of source countries	26	26	26	26	26	26	23	23

Table III(B), page 2

Read:

- Source-country activity restrictions are positively related to aggregate bank outflows.
- Stronger source-country capital regulation, external audit, transparency, and supervisory power tend to increase outflows.
- The source-country results complement the recipient-country inflow results.

Economic message: Stricter home regulation pushes bank capital abroad. This is the “push” side of regulatory arbitrage.

5.5 Figures 1 and 2: Nonparametric changes in regulation and bank flows

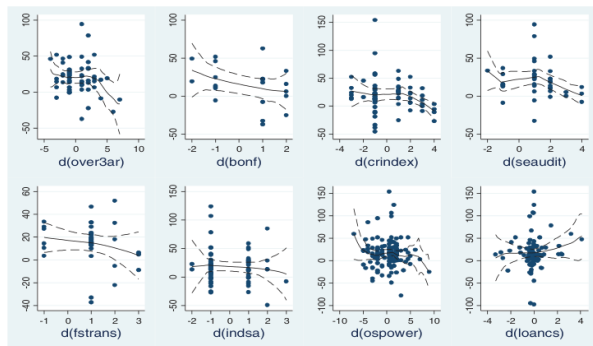


Figure 1. Changes in regulation and changes in credit inflows in recipient countries. This figure presents a nonparametric (kernel-weighted local polynomial smoothing) plot of changes in regulation of recipient countries on the x-axis, and changes in credit inflows on the y-axis. The solid line is fitted credit inflows with the 95% confidence region (Fan and Gijbels (1996)) given in dashed lines. Over3ar is overall activity restrictions for the banking sector, bonf is restrictions on banks owning nonfinancial firms, crindex is capital regulatory index, seaudit is strength of external audit, fstrans is financial statement transparency, indsa is independence of supervisory authority, ospower is official supervisory power, and loancs is loan classification leniency. Among 120 recipient countries in our sample, we find that 114 recipient countries/regions have changed at least one type of regulation during the past decade.

with the changes in regulations of source countries on the x-axis and changes in bank outflow growth on the y-axis. We focus on the three survey years (1999, 2002, and 2005) to measure the regulatory changes. To capture the potential lagged effects of regulatory changes, we use bank flow data in 2001, 2004, and 2007 to measure changes in bank flows.

These plots largely confirm our findings in the regression analyses. In Figure 1, we find a negative relation between changes in various dimensions of bank regulation (i.e., overall activity restrictiveness, restrictions on banks owning

growth

Read:

- Figure 1 plots changes in recipient-country regulation against changes in credit inflows.
- Most panels in Figure 1 show a negative relation: regulation tightening in recipient countries is associated with lower inflow growth.
- Figure 2 plots changes in source-country regulation against changes in credit outflows.
- Most panels in Figure 2 show a positive relation: regulation tightening in source countries is associated with higher outflow growth.

Economic message: The nonparametric plots provide visual evidence that the regression results are not driven solely by functional form or a few large outliers.

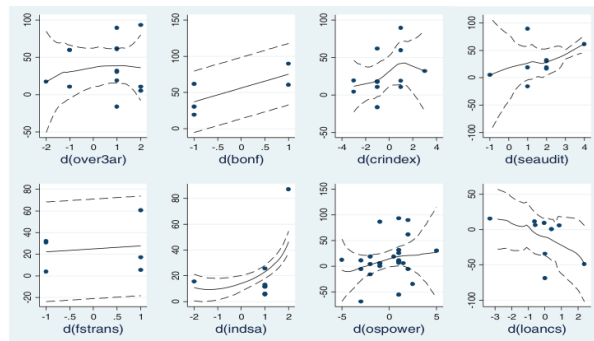


Figure 2. Changes in regulation and changes in credit outflows from source countries. This figure presents a nonparametric (kernel-weighted local polynomial smoothing) plot of changes in regulation of source countries on the x-axis, and changes in credit outflows on the y-axis. The solid line is fitted credit outflows with the 95% confidence region (Fan and Gijbels (1996)) given in dashed lines. Over3ar is overall activity restrictions for the banking sector, bonf is restrictions on banks owning nonfinancial firms, crindex is capital regulatory index, seaudit is strength of external audit, fstrans is financial statement transparency, indsa is independence of supervisory authority, ospower is official supervisory power, and loancs is loan classification leniency. Among 26 source countries in our sample, we find that 21 countries/regions have changed at least one type of regulation during the past decade.

outflows. Taken together, these plots lend credence to our regression findings, provide assurance that the results are not driven by a few outliers, and perhaps provide a clearer picture of the links between regulations and capital market flows.

B. Regulatory Arbitrage and International Bank Flows: Gravity Model

In this section, we combine the “push” and “pull” effects on international bank flows. Figure 2: regulation changes and source-country bank outflow growth

growth

5.6 Table IV: Bilateral gravity model of regulatory arbitrage

Table IV
Regulatory Arbitrage and Bank Flows

The dependent variable is bank flows, which is defined as 100 times the log-difference of total foreign claims ($FCor$) from source country s to recipient country r , that is, $100 \times \Delta \ln(FCor)$. p -values are presented in brackets. The estimations are based on fixed effect OLS regressions for columns (1) to (7) and GMM for column (8). p -values for columns (1) to (7) are computed by the heteroskedasticity-robust standard errors clustered for recipient countries. The GMM estimator and its corresponding p -values for coefficients are based on weighting matrix II of Bekeart, Harvey, and Lundblad (2001, p.477), which is an extended panel-data version of the Newey and West (1987) robust covariance matrix that accommodates serial correlation, cross-country heteroskedasticity, and restricted SUR effects (i.e., the off-diagonal elements of the variance-covariance matrices are restricted to be identical). The lag length of the GMM estimator is set to three due to our limited 10-year sample period (Greene (2008, p.643)). The country-level banking regulatory variables are time varying and are based on three major surveys spanning almost a decade by the World Bank (Barth, Caprio, and Levine (2008)). The values of regulatory variables for the period 1996 to 1999 are taken from the first survey recorded in 1998/1999, for the period 2000 to 2003 are taken from the second survey that assesses the state of regulation as of the end of 2002, and for the period 2004 to 2007 are taken from the third survey that characterizes the environment as of the end of 2005. Detailed variable definitions can be found in Table I. Time-fixed effects, as well as source and recipient country-specific effects, are included in the regressions but not reported. *, **, and *** represent statistical significance at the 10%, 5%, and 1% level, respectively.

	1	2	3	4	5	6	7	8
Overall activity restrictions (source)		0.39 [0.014]**				0.30 [0.005]**	0.26 [0.007]**	0.29 [0.007]**
Overall activity restrictions (recipient)		-0.25 [0.013]**				-0.32 [0.013]**	-0.23 [0.011]**	-0.27 [0.014]**
Restriction on banks owning nonfinancial firms (source)		1.30 [0.071]*				1.56 [0.021]**	1.52 [0.020]**	1.22 [0.022]**
Restriction on banks owning nonfinancial firms (recipient)		-1.97 [0.014]**				-1.41 [0.149]	-1.42 [0.143]	-1.75 [0.116]

(continued)

Table IV, page 1

	1	2	3	4	5	6	7	8
Independence of supervisory authority (source)					0.94 [0.060]**	1.29 [0.140]	1.05 [0.199]	0.87 [0.254]
Independence of supervisory authority (recipient)					-0.71 [0.040]**	-0.55 [0.040]**	-0.83 [0.030]**	-0.68 [0.034]**
Official supervisory power (source)					0.46 [0.036]**	0.61 [0.056]*	0.66 [0.052]*	0.57 [0.070]*
Official supervisory power (recipient)					-0.39 [0.065]*	-0.35 [0.031]**	-0.52 [0.021]**	-0.65 [0.016]**
Loan classification leniency (source)						-0.26 [0.002]**	-0.27 [0.004]**	-0.22 [0.005]**
Loan classification leniency (recipient)					0.43 [0.037]**	0.41 [0.033]**	0.33 [0.039]**	
Creditor rights (source)	-3.68 [0.019]**	-3.74 [0.014]**	-3.03 [0.016]**	-2.74 [0.059]*	-3.31 [0.096]*	-2.68 [0.028]**	-2.40 [0.024]**	-1.83 [0.034]**

(continued)

Table IV, page 3

	1	2	3	4	5	6	7	8
Log income (source)	-1.54 [0.017]**	-1.09 [0.216]	-1.07 [0.170]	-1.72 [0.045]**	-0.98 [0.261]	-1.27 [0.252]	-1.16 [0.219]	-0.93 [0.294]
Log income (recipient)	2.63 [0.050]*	1.88 [0.027]**	1.82 [0.036]**	1.82 [0.031]**	1.21 [0.079]*	2.09 [0.051]*	2.11 [0.071]**	1.50 [0.091]**
Log population (source)	3.75 [0.039]**	2.80 [0.064]*	2.20 [0.269]	2.48 [0.041]**	2.16 [0.108]	1.34 [0.155]	1.60 [0.193]	1.29 [0.240]
Log population (recipient)	2.91 [0.481]	1.67 [0.104]	1.79 [0.220]	1.42 [0.318]	1.35 [0.170]	2.83 [0.030]**	2.48 [0.032]**	2.03 [0.040]**
Common language	2.51 [0.061]**	3.94 [0.068]**	3.99 [0.078]**	4.05 [0.179]	4.06 [0.059]**	4.05 [0.065]**	5.40 [0.007]**	4.30 [0.006]**
Log distance	-1.56 [0.072]**	-1.48 [0.122]	-1.85 [0.017]**	-1.71 [0.111]	-1.66 [0.112]**	-1.64 [0.184]	-1.38 [0.263]	-1.74 [0.258]
Financial liberalization (source)							0.27 [0.162]	0.29 [0.221]
Financial liberalization (recipient)							0.52 [0.011]**	0.42 [0.013]**

(continued)

Table IV, page 5

Table IV—Continued

	1	2	3	4	5	6	7	8
Capital regulatory index (source)			0.59 [0.027]**			0.25 [0.081]*	0.23 [0.078]*	0.18 [0.103]
Capital regulatory index (recipient)			-0.43 [0.072]*			-0.35 [0.033]**	-0.34 [0.026]**	-0.25 [0.031]**
Strength of external audit (source)				0.72 [0.028]**		0.52 [0.158]	0.70 [0.153]	0.61 [0.169]
Strength of external audit (recipient)				-0.26 [0.058]*		-0.47 [0.030]**	-0.65 [0.021]**	-0.53 [0.029]**
Financial statement transparency (source)					1.60 [0.035]**		1.28 [0.027]**	1.85 [0.020]**
Financial statement transparency (recipient)					-0.96 [0.022]**		-0.59 [0.064]*	-0.56 [0.062]*

(continued)

Table IV, page 2

	1	2	3	4	5	6	7	8
Creditor rights (recipient)	4.69 [0.018]**	4.56 [0.020]**	3.94 [0.070]**	4.18 [0.036]**	4.89 [0.048]**	3.39 [0.018]**	3.65 [0.019]**	2.94 [0.024]**
Info sharing (source)	-0.66 [0.011]**	-0.67 [0.016]**	-0.39 [0.061]**	-0.74 [0.008]**	-0.72 [0.011]**	-0.46 [0.043]**	-0.68 [0.040]**	-0.55 [0.063]**
Info sharing (recipient)	0.85 [0.074]**	1.11 [0.021]**	1.01 [0.027]**	1.15 [0.020]**	0.91 [0.063]**	1.12 [0.062]**	1.17 [0.062]**	1.28 [0.034]**
Number of days to enforce contracts (recipient)						0.08 [0.030]**	0.10 [0.040]**	0.08 [0.057]**
Number of days to enforce contracts (source)						-0.14 [0.014]**	-0.15 [0.020]**	-0.12 [0.024]**
Top five bank concentration (recipient)						-2.60 [0.092]*	-3.64 [0.062]**	-2.96 [0.075]*
Government bank ownership (recipient)						-1.09 [0.030]**	-1.67 [0.038]**	-1.37 [0.038]**
Property rights (source)	-2.34 [0.035]**	-2.52 [0.068]**	-2.55 [0.063]**	-3.22 [0.024]**	-2.29 [0.114]	-1.98 [0.035]**	-2.44 [0.031]**	-2.07 [0.042]**
Property rights (recipient)	1.68 [0.031]**	1.91 [0.103]	1.96 [0.040]**	1.95 [0.108]	2.22 [0.067]**	2.50 [0.021]**	1.86 [0.030]**	1.65 [0.037]**

(continued)

Table IV, page 4

	1	2	3	4	5	6	7	8
Growth opportunities (source)							-1.22 [0.027]**	-1.41 [0.037]**
Growth opportunities (recipient)							1.67 [0.029]**	1.16 [0.035]**
Sample period	1996 to 2007	1996 to 2005						
Source	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Recipient	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Country-fixed effects								
Time-fixed effect	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	14,430	13,738	13,790	13,467	13,601	12,836	7,923	7,923
Number of source countries	26	26	26	26	26	26	23	23
Number of recipient countries	120	111	111	108	109	102	70	70
Adj. R ²	0.17	0.18	0.16	0.17	0.18	0.19	0.19	—
Sample period	1996–2007	1996–2007	1996–2007	1996–2007	1996–2007	1996–2007	1996–2005	1996–2005

Table IV, page 6

Read:

- Table IV combines source-country and recipient-country regulation in a bilateral gravity framework.
- Source-country regulatory variables generally have positive coefficients.
- Recipient-country regulatory variables generally have negative coefficients.
- Controls include source and recipient income, population, distance, common language, and institutional variables.
- Results are robust to fixed effects, GMM, and additional growth/liberalization controls.

Economic message: The bilateral gravity model gives the cleanest regulatory-arbitrage interpretation: bank capital flows from stricter regulatory regimes toward lighter regulatory regimes.

5.7 Table V: Instrumental-variable analyses

Table V
Instrumental Variable Analyses

The dependent variable is bank flows, which is defined as 100 times the log-difference of the ratio of total foreign claims (FC_{sr}) from source country s to recipient country r , that is, $100 \times \Delta \ln(FC_{sr})$. The estimation is based on four regressions. First-stage F -test is the test of excluded instrument in the first-stage regression. For Sargan's overidentification test, the null hypothesis is that the instruments used are not correlated with the residuals. Instrumental variables include the experience of banking supervisors, a dummy for central bank as supervisor, Gini coefficients as a measure of income inequality (past five-year moving average), percentage of years since 1776 that a country has been independent, ethnic fractionalization, latitude, and the average regulatory level of other countries in the sample in a specific year. Other control variables include log income (source and recipient), log population (source and recipient), common language, and log distance. In column (6), other control variables also include financial liberalization (source and recipient) and growth opportunity (source and recipient). Time-fixed effects, as well as source and recipient country-specific effects are included in the regressions but not reported. Detailed variable definitions can be found in Table I. p -values are computed by the heteroskedasticity-robust standard errors clustered for recipient countries and are presented in brackets. *, **, and *** represent statistical significance at the 10%, 5%, and 1% level, respectively.

	1	2	3	4	5	6
Overall activity	0.86				0.60	0.53
restrictions (source)	[0.003]***				[0.000]***	[0.000]***
Overall activity	-0.43				-0.57	-0.46
restrictions (recipient)	[0.013]**				[0.016]**	[0.015]**
Restriction on banks	2.55				2.50	2.73
owning nonfinancial firms	[0.069]*				[0.020]**	[0.015]**
(source)						
Restriction on banks	-2.82				-2.66	-2.08
owning nonfinancial firms	[0.012]**				[0.126]	[0.114]
(recipient)						
Capital regulatory index		0.87			0.43	0.46
(source)		[0.027]**			[0.092]*	[0.077]*
Capital regulatory index		-0.83			-0.73	-0.98
(recipient)		[0.059]*			[0.034]**	[0.018]**
Strength of external			1.40		0.73	1.18
audit (source)			[0.024]**		[0.484]	[0.241]
Strength of external			-0.57		-0.94	-1.20
audit (recipient)			[0.062]*		[0.020]**	[0.009]***
Financial statement			3.09		2.60	3.66
transparency (source)			[0.031]**		[0.004]***	[0.012]*
Financial statement			-1.66		-1.09	-1.17
transparency (recipient)			[0.028]**		[0.021]**	[0.059]*
Independence of				1.66	2.23	1.48
supervisory authority				[0.053]*	[0.047]**	[0.302]
(source)						
Independence of				-1.16	-0.87	-1.33
supervisory authority				[0.041]**	[0.041]**	[0.025]**
(recipient)						
Official supervisory				0.87	1.31	0.95
power (source)				[0.038]**	[0.032]**	[0.036]**
Official supervisory				-0.67	-0.94	-1.06
power (recipient)				[0.060]*	[0.029]**	[0.023]**
Loan classification					-0.45	-0.38
leniency (source)					[0.000]***	[0.015]**

(continued)

Table V—Continued

	1	2	3	4	5	6
Loan classification					0.66	0.56
leniency (recipient)					[0.014]**	[0.027]**
Creditor rights (source)	-4.86	-3.58	-2.81	-3.16	-2.44	-3.62
	[0.007]***	[0.016]**	[0.055]*	[0.102]	[0.020]**	[0.014]**
Creditor rights (recipient)	4.56	3.86	4.16	4.54	3.22	3.61
	[0.020]**	[0.069]*	[0.034]**	[0.044]**	[0.014]**	[0.017]**
Info sharing (source)	-0.70	-0.39	-0.74	-0.67	-0.47	-0.06
	[0.015]**	[0.060]*	[0.008]***	[0.015]**	[0.045]**	[0.059]*
Info sharing (recipient)	1.12	1.00	1.15	0.94	1.23	1.42
	[0.021]**	[0.027]**	[0.018]**	[0.061]*	[0.046]**	[0.045]**
Property rights (source)	-2.87	-2.56	-3.25	-2.47	-1.73	-2.44
	[0.054]*	[0.061]*	[0.022]**	[0.125]	[0.029]**	[0.039]**
Property rights	1.89	1.95	1.94	2.23	2.32	1.87
(recipient)	[0.105]	[0.039]**	[0.106]	[0.063]*	[0.022]**	[0.031]**
Sample period		1996 to 2007			1996 to 2005	
Other control variables	Yes	Yes	Yes	Yes	Yes	Yes
Source country-fixed	Yes	Yes	Yes	Yes	Yes	Yes
Recipient country-fixed	Yes	Yes	Yes	Yes	Yes	Yes
effects						
Time-fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
First-stage F -test (p -value)	0.000	0.000	0.000	0.000	0.000	0.000
Sargan's overidentification	0.24	0.23	0.16	0.21	0.33	0.19
test (p -value)						
Observations	13,738	13,790	13,467	13,601	12,936	7,923
Number of recipient	26	26	26	26	26	23
countries						
Number of source countries	111	111	108	109	102	70
Adj. R^2	0.26	0.25	0.23	0.24	0.26	0.27
Sample period	1996–2007	1996–2007	1996–2007	1996–2007	1996–2007	1996–2005

heavily regulated to markets that are lightly regulated. Other controls also yield qualitatively similar results.

Furthermore, the instrumental variable coefficients are somewhat larger than the OLS coefficients, indicating potential measurement error in the original results that would “attenuate” the coefficient estimate toward zero (Rajan and Subramanian (2008), Barth et al., (2009)). Following the literature (e.g.,

Table V, page 1

Table V, page 2

Read:

- IV specifications continue to show positive source-regulation effects and negative recipient-regulation effects.
- The instruments are designed to capture plausibly exogenous variation in regulatory environments.
- The first-stage and overidentification diagnostics are reported with the estimates.

Economic message: Table V addresses the concern that regulation may be endogenous to bank flows. The main pattern survives.

5.8 Table VI: Split-sample estimation across time and institutional quality

Table VI
Split Sample Estimation: Regulation Arbitrage Activities over Time and across Quality of Institutions

The dependent variable is bank flow, which is defined as 100 times the log-difference of the total foreign claims (FCr) from source country s to recipient country r , that is, $100 \times \Delta \ln(FCr)$. The estimation is based on fixed effect OLS regressions. The country-level banking regulatory variables are time varying and are based on three major surveys spanning almost a decade by the World Bank (Barth, Caprio, and Levine (2008)). The values of regulatory variables for the period 1996 to 1999 are taken from the first survey recorded in 1998/1999, for the period 2000 to 2003 are taken from the second survey that assesses the state of regulation as of the end of 2002, and for the period 2004 to 2007 are taken from the third survey that characterizes the environment as of the end of 2005. Detailed variable definitions can be found in Table 1. Other control variables include log income (gpi), log population (gpi), common language, and log distance. Time-fixed effects, as well as source and recipient country-specific effects are included in the regressions but not reported. p -values are computed using heteroskedasticity-robust standard errors clustered for recipient countries and are presented in brackets. ":", "*", and "***" represent statistical significance at the 10%, 5%, and 1% level, respectively. To define high-/low-quality institution recipient countries in terms of the indicators creditor rights, depth of information sharing (info share), and property rights, we first calculate the average institution quality of each indicator for each country over time. If the average institution quality of an indicator for a country is equal to or greater than the sample median level, the country is defined as a high-quality institution country for this particular indicator. The remaining countries are defined as low-quality institution countries for this indicator. Next we combine the above three individual institution quality indicators to obtain an aggregate measure of overall institution quality for each country. For each individual indicator, we assign a value of one to a high-quality country and zero to a low-quality country. We then sum the scores of each country across the three indicators to obtain the overall measure of a country's institution quality. The value of this aggregate measure goes from zero (lowest institution quality country) to three (highest institution quality country). If the aggregate measure of a country is equal to or greater than the sample median level, the country is defined as an overall high-quality institution country. The remaining countries are defined as low-quality institution countries.

	1	2	3	4	5	6	7	8	9	10	11	12
	Split Sample over Time	Bank Flow from Adv. to Emerging/ Dev. Economies	Bank Flow from Adv. to High Creditor Rights	Bank Flow from Adv. to Low Property Rights	Bank Flow from Adv. to High-Quality Institution	Bank Flow from Adv. to Low-Quality Institution	Bank Flow from Adv. to High-Quality Institution	Bank Flow from Adv. to Low-Quality Institution	Bank Flow from Adv. to High-Quality Institution	Bank Flow from Adv. to Low-Quality Institution	Bank Flow from Adv. to High-Quality Institution	Bank Flow from Adv. to Low-Quality Institution
Sample Period	1996 to 2002 to 2007	1996 to 2002 to 2007	1996 to 2002 to 2007	1996 to 2002 to 2007	1996 to 2002 to 2007	1996 to 2002 to 2007	1996 to 2002 to 2007	1996 to 2002 to 2007	1996 to 2002 to 2007	1996 to 2002 to 2007	1996 to 2002 to 2007	1996 to 2002 to 2007
Overall	0.61	0.70	1.13	0.27	1.03	0.61	0.76	0.25	2.12	0.15	1.31	0.12
activity	[0.028]**	[0.018]**	[0.004]***	[0.032]**	[0.005]***	[0.048]	[0.021]**	[0.037]**	[0.006]***	[0.032]	[0.006]***	[0.040]
restrictions												
(gpi)												

(continued)

Table VI—Continued

	1	2	3	4	5	6	7	8	9	10	11	12
	Split Sample over Time	Bank Flow from Adv. to Emerging/ Dev. Economies	Bank Flow from Adv. to High Creditor Rights	Bank Flow from Adv. to Low Property Rights	Bank Flow from Adv. to High-Quality Institution	Bank Flow from Adv. to Low-Quality Institution	Bank Flow from Adv. to High-Quality Institution	Bank Flow from Adv. to Low-Quality Institution	Bank Flow from Adv. to High-Quality Institution	Bank Flow from Adv. to Low-Quality Institution	Bank Flow from Adv. to High-Quality Institution	Bank Flow from Adv. to Low-Quality Institution
Restriction on banks owning nonfinancial firms (gpi)	1.41	1.48	1.91	0.96	1.83	0.43	0.54	0.43	1.63	0.62	1.93	0.92
	[0.031]**	[0.024]**	[0.012]**	[0.039]**	[0.019]**	[0.069]**	[0.027]**	[0.036]**	[0.024]**	[0.031]**	[0.004]***	[0.018]**
Capital regulatory index (gpi)	0.38	0.56	0.32	0.28	0.50	0.14	0.61	0.34	0.67	0.12	0.38	0.21
	[0.031]**	[0.011]**	[0.034]**	[0.039]**	[0.012]**	[0.035]**	[0.008]**	[0.008]**	[0.002]**	[0.031]**	[0.009]**	[0.034]**
Strength of external audit (gpi)	0.93	1.73	2.06	0.12	1.24	0.32	1.54	0.63	2.05	0.23	1.82	0.22
	[0.084]**	[0.021]**	[0.008]**	[0.216]	[0.026]**	[0.236]	[0.026]**	[0.298]	[0.002]**	[0.222]	[0.014]**	[0.232]
Fin statement transparency (gpi)	1.39	1.77	0.69	0.27	0.49	0.40	0.71	0.61	0.24	0.11	0.43	0.32
	[0.033]**	[0.006]***	[0.025]**	[0.039]**	[0.033]**	[0.042]**	[0.030]**	[0.040]**	[0.029]**	[0.081]**	[0.029]**	[0.034]**
Independence of supervisory authority: overall (gpi)	0.77	1.33	0.86	0.44	0.54	0.35	0.65	0.25	1.92	1.40	0.54	0.23
	[0.036]**	[0.028]**	[0.027]**	[0.066]**	[0.027]**	[0.031]**	[0.038]**	[0.061]**	[0.011]**	[0.011]**	[0.026]**	[0.056]**
Official supervisory power (gpi)	0.18	1.28	0.39	0.30	0.23	0.15	0.39	0.12	0.31	0.28	0.25	0.14
	[0.163]	[0.001]***	[0.014]**	[0.041]**	[0.035]**	[0.071]**	[0.026]**	[0.073]**	[0.028]**	[0.051]**	[0.012]**	[0.077]**
Loan classification (gpi)	-0.63	-1.26	-2.19	-0.93	-0.23	-0.21	-1.62	-0.64	-0.60	-0.58	-1.34	-0.92
	[0.059]**	[0.035]**	[0.006]**	[0.028]**	[0.028]**	[0.083]**	[0.020]**	[0.045]**	[0.034]**	[0.058]**	[0.019]**	[0.022]**
Creditor rights (gpi)	-2.99	-3.93	-2.10	-2.49	-0.33	-2.25	-0.52	-3.64	-0.87	-3.45	-0.13	-5.07
	[0.029]**	[0.014]**	[0.037]**	[0.036]**	[0.069]**	[0.034]**	[0.072]**	[0.026]**	[0.152]	[0.034]**	[0.192]	[0.012]**
Info share (gpi)	-0.94	-1.49	-0.65	-1.87	-0.54	-1.24	-1.01	-1.24	-0.12	-2.90	-0.52	-1.75
	[0.052]**	[0.026]**	[0.073]**	[0.097]**	[0.013]**	[0.038]**	[0.018]**	[0.096]**	[0.008]**	[0.086]**	[0.002]**	[0.062]**
Property rights (gpi)	-2.28	-2.46	-1.91	-2.25	-0.39	-0.86	-0.67	-1.29	-2.01	-2.87	-1.44	-1.62
	[0.017]**	[0.176]	[0.012]**	[0.007]**	[0.073]**	[0.052]**	[0.128]	[0.040]**	[0.014]**	[0.000]**	[0.027]**	[0.004]**
Other controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

(continued)

Table VI, page 1

Table VI, page 2

Table VI—Continued

	1	2	3	4	5	6	7	8	9	10	11	12
	Split Sample over Time	Bank Flow from Adv. to Emerging/ Dev. Economies	Bank Flow from Adv. to High Creditor Rights	Bank Flow from Adv. to Low Property Rights	Bank Flow from Adv. to High-Quality Institution	Bank Flow from Adv. to Low-Quality Institution	Bank Flow from Adv. to High-Quality Institution	Bank Flow from Adv. to Low-Quality Institution	Bank Flow from Adv. to High-Quality Institution	Bank Flow from Adv. to Low-Quality Institution	Bank Flow from Adv. to High-Quality Institution	Bank Flow from Adv. to Low-Quality Institution
Source country-fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Recipient country fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Time-fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	5,446	7,640	4,150	8,398	7,837	5,249	6,652	6,434	6,675	6,411	6,828	6,258
Number of source countries	19	26	21	19	26	26	26	24	26	24	26	25
Number of recipient countries	103	103	29	74	63	47	45	58	45	58	47	58
Adj. R^2	0.14	0.15	0.14	0.16	0.15	0.14	0.13	0.14	0.13	0.14	0.13	0.14
Sample period			1996–2007	1996–2007	1996–2007	1996–2007	1996–2007	1996–2007	1996–2007	1996–2007	1996–2007	1996–2007

Table VI, page 3

Read:

- The regulatory-arbitrage effect is stronger in later years than in the earlier period.
- Effects are stronger for high-creditor-rights countries than for low-creditor-rights countries.
- Effects are stronger when information sharing, property rights, or overall institutional quality are high.
- Developed and institutionally strong countries are especially important destinations for regulatory arbitrage.

Economic message: Lax regulation alone is not enough. Bank capital is more likely to chase regulatory gaps when the recipient country also has strong legal and institutional foundations.

5.9 Table VII: Bank-level foreign affiliates and regulatory arbitrage

Table VII
Probit Estimation: Bank's Foreign Affiliates and Regulatory Arbitrage

The dependent variable for columns (1) to (5) and (7) is a dummy that takes the value of one if a bank headquartered in source country s has a foreign subsidiary in recipient country r , and zero otherwise, in 2008. The dependent variable for column (6) is a dummy that takes the value of one if the bank headquartered in source country s has either a foreign subsidiary or a foreign branch in recipient country r , and zero otherwise, in 2008. The estimation is via a probit model. The explanatory variables are for 2007. Bank size and net income are the three-year average values over 2005 to 2007. The coefficient estimates are transformed to represent the marginal effects evaluated at the mean of the independent variables from the interval regressions. The marginal effect of a dummy variable is calculated as the discrete change in the expected value of the dependent variable as the dummy variable changes from zero to one. Other control variables include log income (source and recipient), log population (source and recipient), common language, and log distance. p -values are computed using heteroskedasticity-robust standard errors clustered for recipient countries and are presented in brackets. *, **, and *** represent statistical significance at the 10%, 5%, and 1% level respectively.

	1	2	3	4	5	6	7
Overall activity restrictions (source)	0.029 [0.067]**				0.028 [0.061]**	0.030 [0.012]**	0.043 [0.023]**
Overall activity restrictions (recipient)	-0.018 [0.027]**				-0.012 [0.023]**	-0.014 [0.014]**	
Restriction on banks owning nonfinancial firms (source)	0.063 [0.024]**				0.064 [0.074]**	0.065 [0.031]**	0.069 [0.031]**
Restriction on banks owning nonfinancial firms (recipient)	-0.049 [0.029]**				-0.041 [0.020]**	-0.036 [0.017]**	-0.049 [0.028]**
Capital regulatory index (source)		0.027 [0.018]**			0.027 [0.015]**	0.021 [0.021]**	0.049 [0.026]**
Capital regulatory index (recipient)		-0.011 [0.014]**			-0.012 [0.016]**	-0.014 [0.029]**	-0.027 [0.012]**
Strength of external audit (source)			0.040 [0.032]**		0.044 [0.038]**	0.053 [0.079]**	0.056 [0.013]**
Strength of external audit (recipient)			-0.032 [0.128]		-0.031 [0.025]**	-0.047 [0.031]**	-0.048 [0.026]**
Financial statement transparency (source)			0.035 [0.011]**		0.030 [0.030]**	0.020 [0.029]**	0.020 [0.015]**
Financial statement transparency (recipient)			-0.018 [0.019]**		-0.024 [0.024]**	-0.016 [0.019]**	-0.032 [0.028]**

(continued)

Table VII—Continued

	1	2	3	4	5	6	7
Independence of supervisory authority (source)				0.057 [0.017]**	0.046 [0.012]**	0.051 [0.033]**	0.049 [0.009]**
Independence of supervisory authority (recipient)				-0.023 [0.015]**	-0.025 [0.031]**	-0.027 [0.029]**	-0.029 [0.023]**
Official supervisory power (source)				0.036 [0.011]**	0.037 [0.018]**	0.038 [0.053]**	0.039 [0.011]**
Official supervisory power (recipient)				-0.028 [0.051]*	-0.025 [0.016]**	-0.020 [0.014]**	-0.049 [0.031]**
Loan classification leniency (source)					-0.044 [0.060]**	-0.041 [0.035]**	-0.057 [0.038]**
Loan classification leniency (recipient)					0.025 [0.030]**	0.024 [0.058]**	0.021 [0.013]**
Government bank ownership (recipient)					-0.361 [0.077]**	-0.376 [0.034]**	-0.355 [0.034]**
Top five bank concentration (all banks) (recipient)					-0.208 [0.016]**	-0.204 [0.016]**	-0.234 [0.035]**
Top five bank concentration (excl. foreign banks) (recipient)						-0.257 [0.016]**	
Bank size		0.016 [0.130]	0.015 [0.178]	0.014 [0.118]	0.016 [0.133]	0.017 [0.042]**	0.016 [0.168]
Bank net income		0.036 [0.033]**	0.031 [0.088]*	0.030 [0.060]*	0.023 [0.113]	0.038 [0.047]**	0.057 [0.186]
Creditor rights (source)			-0.039 [0.058]**	-0.034 [0.060]**	-0.035 [0.058]**	-0.032 [0.022]**	-0.039 [0.026]**
Creditor rights (recipient)			0.024 [0.083]**	0.026 [0.060]**	0.024 [0.048]**	0.018 [0.056]**	0.025 [0.063]**
Info sharing (source)			-0.031 [0.044]**	-0.032 [0.024]**	-0.032 [0.036]**	-0.034 [0.078]**	-0.038 [0.021]**
Info sharing (recipient)						-0.038 [0.069]**	-0.030 [0.020]**

(continued)

Table VII, page 1

Table VII, page 2

Table VII—Continued

	1	2	3	4	5	6	7
Info sharing (recipient)	0.015 [0.087]**	0.016 [0.008]**	0.021 [0.006]**	0.014 [0.086]**	0.019 [0.069]**	0.024 [0.037]**	0.032 [0.011]**
Property rights (source)	-0.024 [0.053]**	-0.025 [0.043]**	-0.020 [0.008]**	-0.028 [0.051]**	-0.037 [0.084]**	-0.037 [0.003]**	-0.027 [0.078]**
Property rights (recipient)	0.019 [0.062]**	0.022 [0.036]**	0.015 [0.029]**	0.024 [0.093]**	0.029 [0.043]**	0.021 [0.068]**	0.026 [0.019]**
Financial liberalization (source)							0.039 [0.319]
Financial liberalization (recipient)							0.044 [0.028]**
Growth opportunities (source)							-0.317 [0.016]**
Growth opportunities (recipient)							0.712 [0.018]**
Source countries	26	26	26	26	26	26	24
Recipient countries	111	111	108	109	103	103	70
Banks	301	301	301	301	301	301	284
Other control variables	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	33,110	33,110	32,207	32,508	30,702	30,702	19,596

Table VII, page 3

Read:

- The dependent variables are foreign subsidiary and foreign presence.
- Banks headquartered in more regulated countries are more likely to establish foreign affiliates.
- Host-country regulation has the opposite sign: stricter host-country regulation reduces the likelihood of foreign subsidiaries or branches.
- Results hold after adding bank size, profitability, net income, and country controls.

Economic message: Regulatory arbitrage affects not only financial claims but also the organizational geography of multinational banks.

6 Short summary

- The paper studies whether regulatory differences influence cross-border bank flows.

- BIS data provide bilateral foreign claims from 26 source countries to 120 recipient countries over 1996–2007.
- Bank-flow growth is measured as the log change in foreign claims.
- Bank regulation is measured using Barth, Caprio, and Levine survey data.
- Recipient-country regulatory strictness is negatively related to bank inflows.
- Source-country regulatory strictness is positively related to bank outflows.
- Bilateral gravity models confirm the push-pull pattern.
- Nonparametric plots show that changes in regulation line up with changes in flows.
- IV regressions support the baseline results.
- Regulatory arbitrage is stronger when destination countries are developed and institutionally strong.
- Bank-level probit tests show that banks from regulated countries are more likely to expand into less regulated host countries.
- The main policy implication is that purely domestic bank regulation can be undermined by cross-border movement of bank capital.

7 Conclusion

7.1 Core conclusion

Houston, Lin, and Ma provide evidence that international bank flows respond to cross-country regulatory gaps. Bank capital tends to flow out of stricter regulatory environments and into less regulated ones. This supports the regulatory-arbitrage concern that national regulation can be weakened when banks can shift activity internationally.

7.2 Economic intuition

Regulation imposes constraints and costs. If banks can serve customers or hold claims abroad, they may move activity toward countries where constraints are lighter. However, they do not move purely toward the weakest regulatory regimes. Legal and institutional quality still matters because banks need enforceable claims, creditor protection, and property rights.

7.3 Takeaway

The paper’s main message is that regulatory arbitrage is real but conditional. Lax regulation attracts bank capital most strongly when it is paired with strong institutions. This means global regulatory coordination is important, but full uniformity is not the only issue: institutional quality also shapes where bank capital moves.